ASSIGNMENT

SHORT ANSWER TEST

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computer
vision

The problem is insufficient sampling of the image. which results in fewer pixels and makes small objects unclear. increasing the sampling rate / spatial resolution by using a higher-resolution camera can capture more pixels. This improves the representation and visibility of small objects.

2. A suitable technique is image enhancement using methods like histogram equalization, combined with a image-processing techniques like noise reduction. filter if required. Histogram equalization improves the contrast of dark regions, while filtering reduces the noise. This make objects more visible in night image.

3. Quantization converts continuous intensity values into a finite number of discrete intensity values. For example, a image can use 256 levels ranging from 0-255. This allows the continuous image signal to be represented digitally and stored and processed by a computer.

4. An industrial camera captures a product with uneven lighting that makes some defects difficult to see. To solve this problem we can use contrast enhancement techniques because it helps the industrial cameras to capture defects even in the uneven lighting which helps in improving the defect visibility.

5. Image segmentation can divide the image into regions such as road, vehicles, animals and pedestrians

Semantic or instance segmentation can be used to label these regions. So, this helps the autonomous self driving cars to understand the surroundings and make safe decisions.

6. A factory inspection system needs to identify cracks on manufactured components. Edge detection such as Canny edge technique can be applied to identify sharp intensity changes caused by cracks. The detected edges highlight the boundaries and shapes of cracks. This allows the inspection system to locate and measure defects.

7. A suitable technique is Guassian filtering or median filtering, deping on the type of noise median filtering is particularly effective for salt-and-pepper noise while preserving important edges. Filtering reduces unwanted noise and makes the tumor easy to identify.

8. Object detection algorithms such as YOLO can identify vehicles in each video frame. Object tracking can then maintain the identity and movement of each detected vehicle. This helps in traffic monitoring, vehicle counting, congestion detection, and accident analysis.

9. The possible cause is motion blur, which occurs when the object or camera moves while the image is being captured. Deblurring or image restoration

techniques can be applied to improve the image quality and to reduce noise / blur.

An industrial vision system must distinguish defective products from good products based on their appearance. Image classification can classify the products as defective and good products. based on their visual features. A trained CNN model can learn patterns from product images.